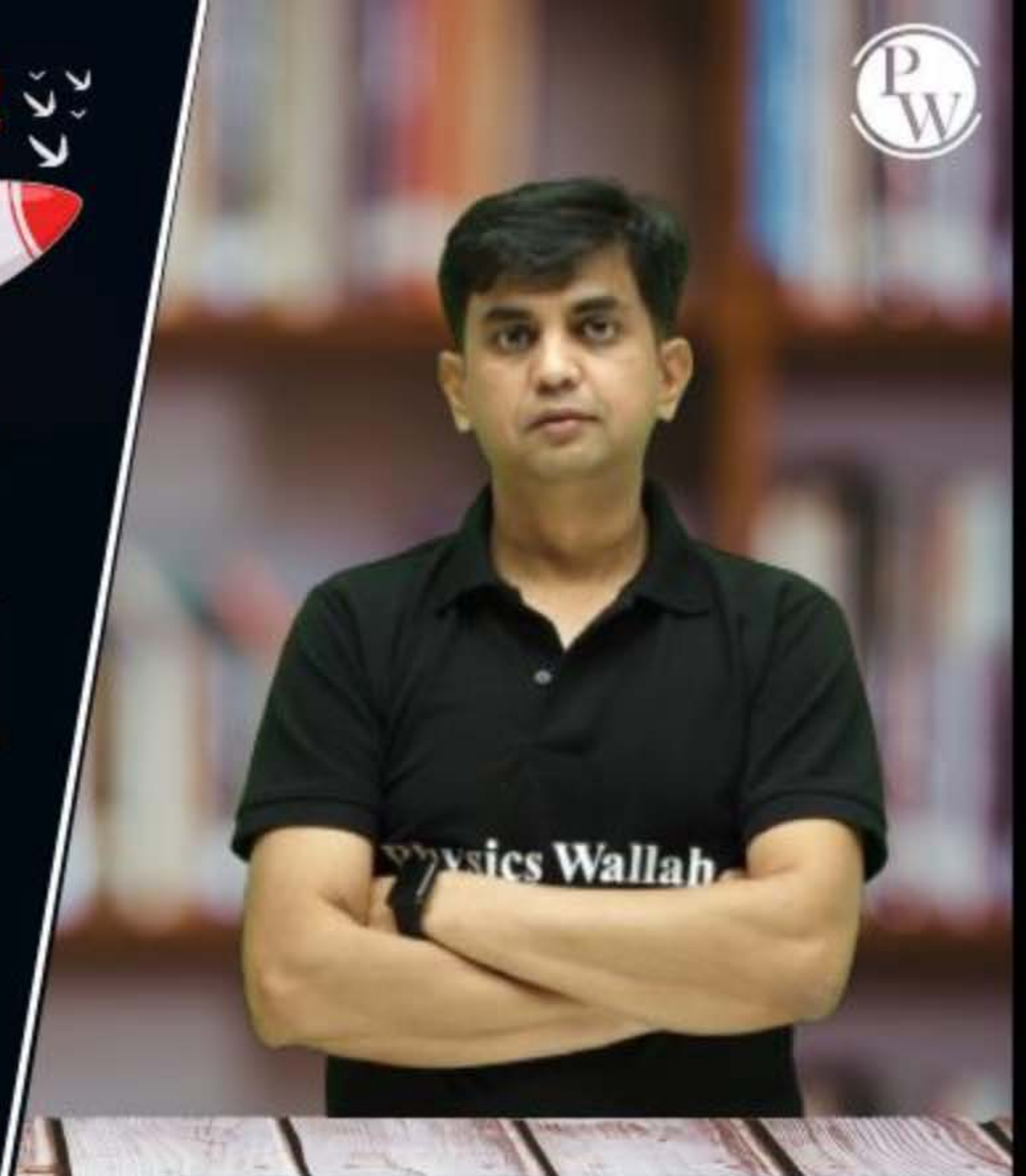
CS & IT ENGINEERING

Computer Network

Flow Control



Lecture No. - 02

By - Abhishek Sir

Recap of Previous Lecture

Topic Network Delays

Topics to be Covered

Topic: Flow Control Protocols

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider two hosts X and Y, connected by a single direct link of $\boxed{\text{rate}}$ 10^6 bits/sec. The distance between the two hosts is $\boxed{10{,}000 \text{ km}}$ and the propagation speed along the link is $\boxed{2 \times 10^8 \text{ m/s.}}$ Hosts X send a file of 50,000 bytes as one large message to hosts Y continuously. Let the transmission and propagation delays be $\boxed{p \text{ milliseconds}}$ and $\boxed{q \text{ milliseconds,}}$ respectively. Then the vales of p and q are:

[GATE-2017, Set-2, MCQ, 2-Mark]

~~(A)~~ p = 50 and q = 100

~~(B)~~ p = 50 and q = 400

~~(C)~~ p = 100 and q = 50

(D) p = 400 and $\boxed{q = 50}$

$\boxed{\text{Ans: D}}$

Solution:-

$\underbrace{\text{Packet Size}}$ = $\boxed{50{,}000 \text{ bytes}}$ = $\boxed{4 * 10^5 \text{ bits}}$

$\underbrace{\text{Bandwidth}}$ = $\boxed{10^6 \text{ bits / sec}}$

$$p = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{4 * 10^5 \cancel{\text{bits}}}{10^6 \cancel{\text{bits}} / \text{sec}} = \frac{4 * 10^5}{10^3} * 10^{-3} \text{ sec}$$

$$= 400 \text{ ms}$$

$\underbrace{\text{Distance}} = \boxed{10{,}000\ \text{Km}} = 10^7\ \text{m}$

$\underbrace{\text{Signal Speed}} = \boxed{2 \times 10^8\ \text{m/s}}$

$$q = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{10^7\ \cancel{m}}{2 \times 10^8\ \cancel{m}/s} = \frac{10 * 10^6}{2 * 10^5} * 10^{-3}\ \text{sec}$$

$$= 50\ \text{ms}$$

#Q. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3 x10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is _______.

[GATE-2022, NAT, 2-Marks]

Solution:-

Packet Size = $\boxed{1000 \text{ bytes}}$ = $\boxed{8 * 10^3 \text{ bits}}$

Bandwidth = 100 Mbps = $\boxed{10^8 \text{ bits / sec}}$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{10^8 \text{ bits / sec}} = \frac{8*10^3}{10^5} * 10^{-3} \text{ sec}$$

$$= \frac{8}{100} \text{ ms} = 0.08 \text{ ms}$$

Distance $\quad = \boxed{2100 \text{ Km}} = 21 * 10^5$ m

Signal Speed $= \boxed{3 \times 10^8 \text{ m/s}}$

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{21 * 10^5 \text{ m}}{3 \times 10^8 \text{ m/s}} = \frac{21 * 10^5}{3 * 10^5} * 10^{-3} \text{ sec}$$

$$= 7 \text{ ms}$$

End-to-end delay $= (t_x + t_p)$

$$= (7 + 0.08) \text{ ms}$$

$$= 7.08 \text{ ms}$$

$\boxed{\text{IIT KGP Ans: } 7.07 \text{ to } 7.09}$

Topic : Flow Control

→ Synchronization between transmitter and receiver to control the flow

→ Flow Control Protocols :

1. Noiseless Channel
   - → No chance of error in the link
   - → No any 'error detection' required
2. Noisy Channel
   - → Chance of error in the link
   - → Need of 'error detection' technique

Topic : Flow Control

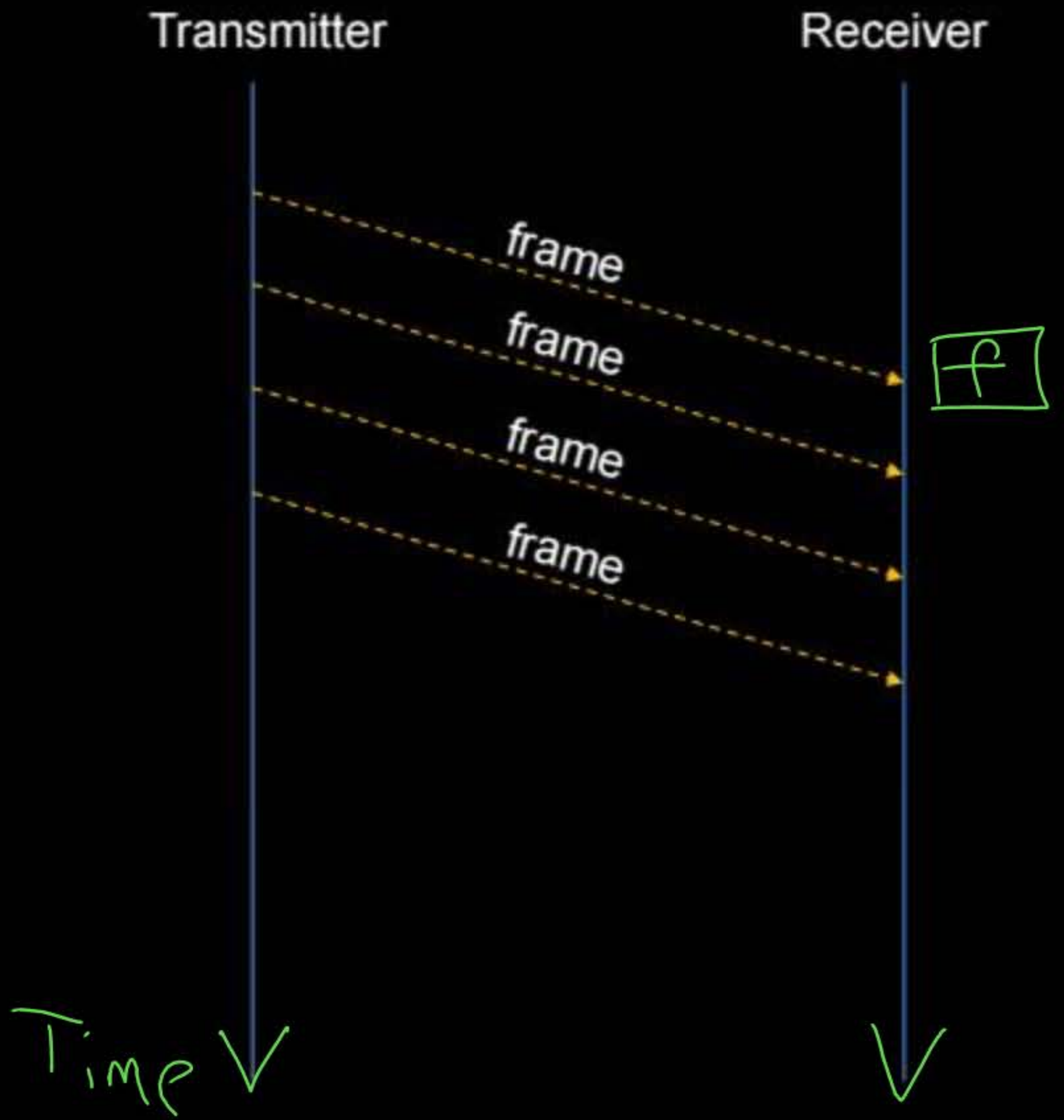
→ Flow Control Protocols for Noiseless Channel :

1. Simplest
2. Stop-and-Wait

Topic : Simplest

→ No any Flow and Error Control

→ Like Simplex Communication (No any ACK)



Topic : Stop-and-Wait

→ Add Flow Control in Simplest

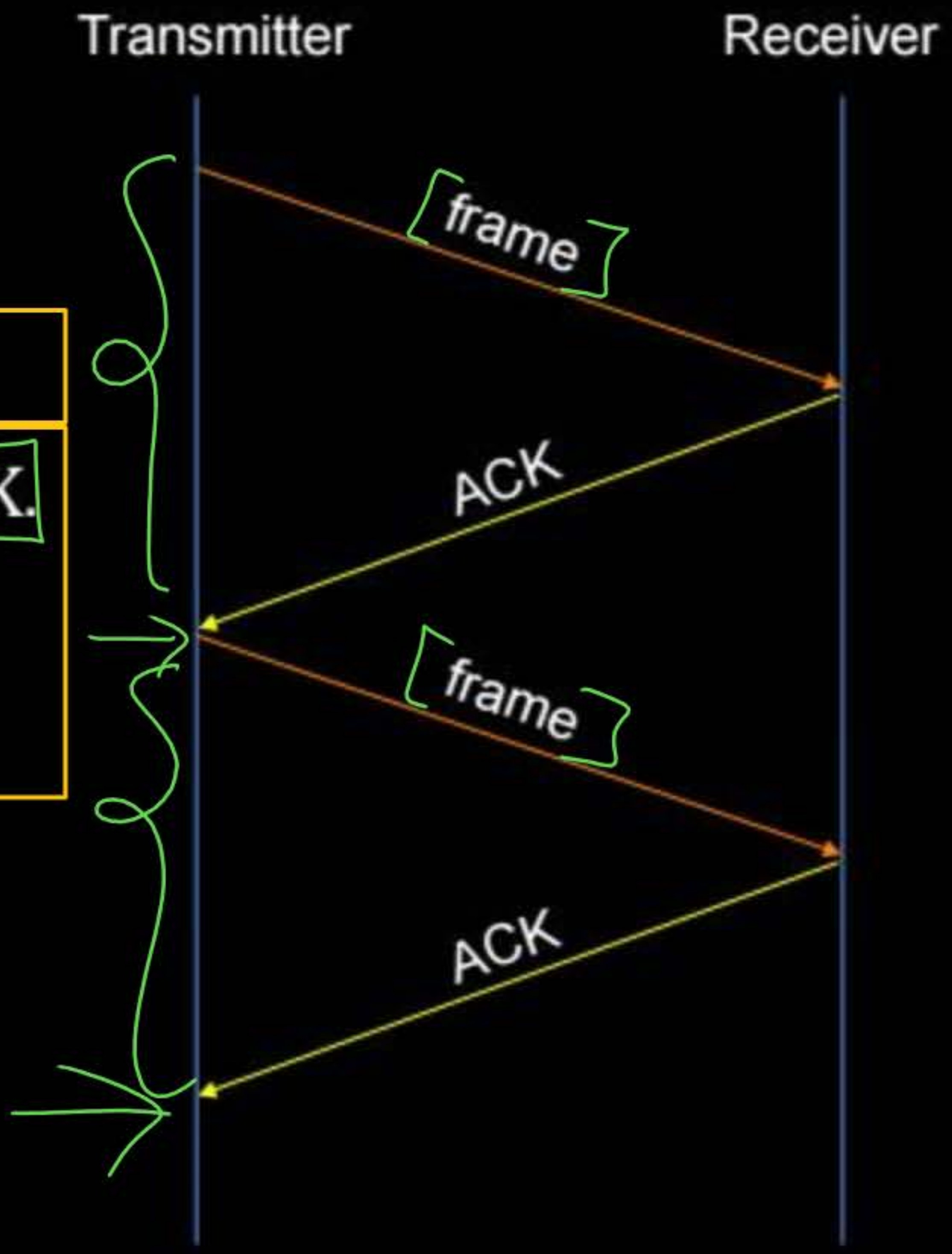
→ No any Error Control

→ Receiver send ACK for every received frame

→ Transmitter transmit one frame and wait for an ACK.

→ Transmitter transmit next frame only after receiving ACK of transmitted frame

→ Like Half Duplex Communication



Topic : Flow Control

→ Flow Control Protocols for Noisy Channel :

1. Stop-and-Wait ARQ
2. Sliding Window ARQ

    2.1 Go-Back-N ARQ

    2.2 Selective-Repeat ARQ

ARQ : Automatic Repeat Request

Topic : Stop-and-Wait ARQ

→ Transmitter transmit one frame and wait for an ACK

→ Receiver send ACK (positive ACK) for every successfully received frame

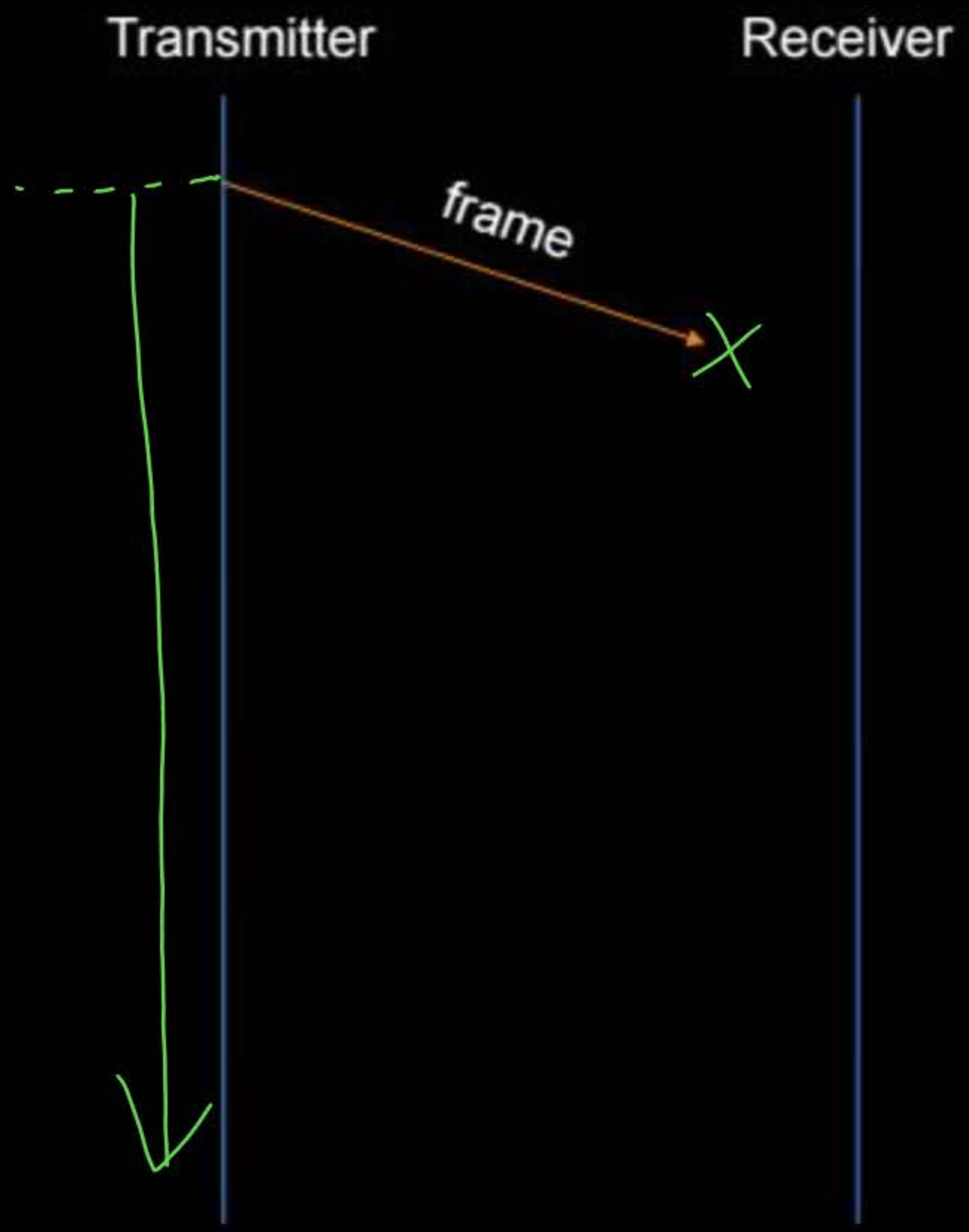
→ Transmitter transmit next frame
only after receiving ACK of transmitted frame

Topic : Stop-and-Wait ARQ

Case I :

→ Either frame or $\boxed{\text{ACK}}$ gets lost in the channel

→ Transmitter may goes in indefinite wait for ACK



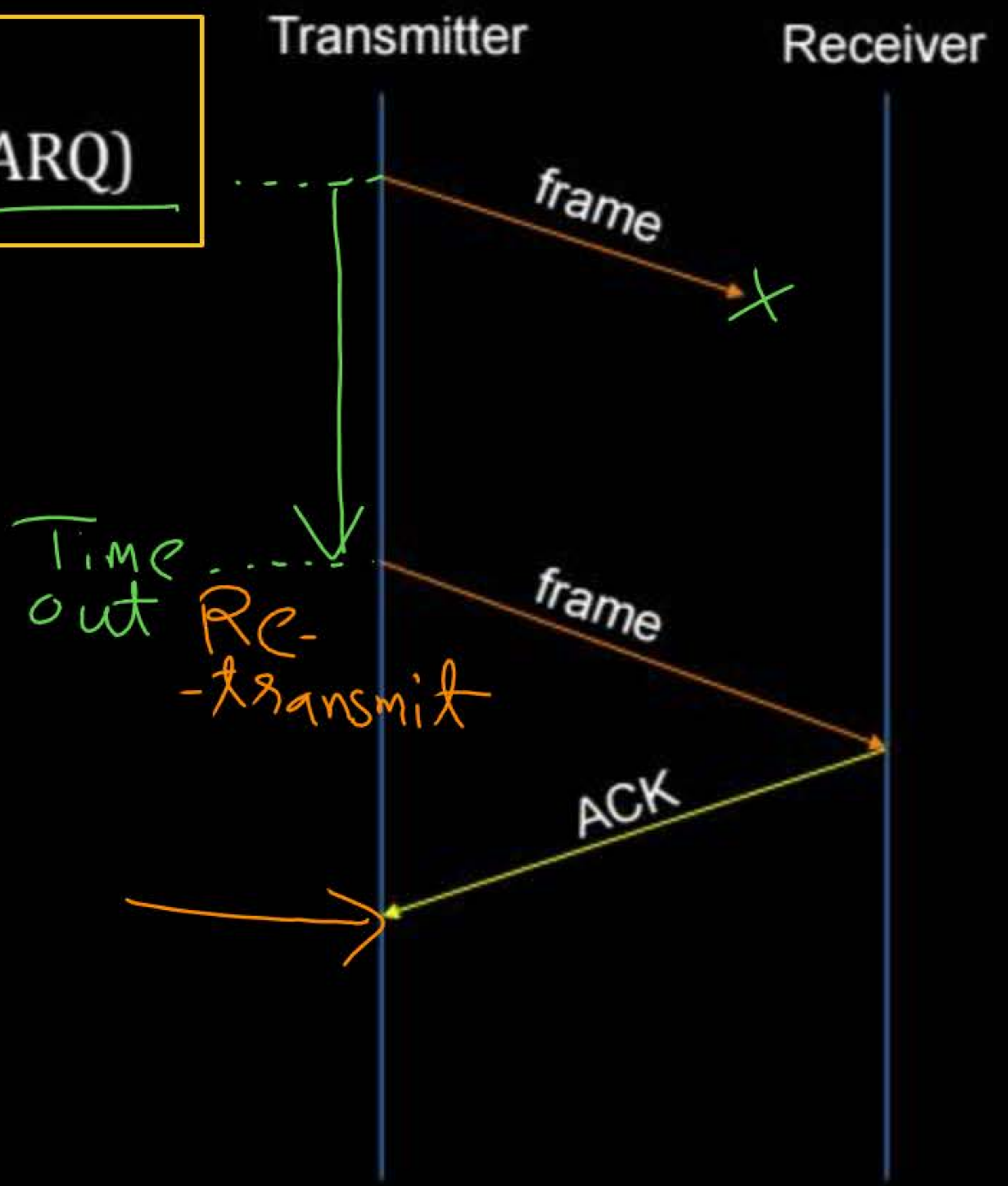
Topic : Stop-and-Wait ARQ

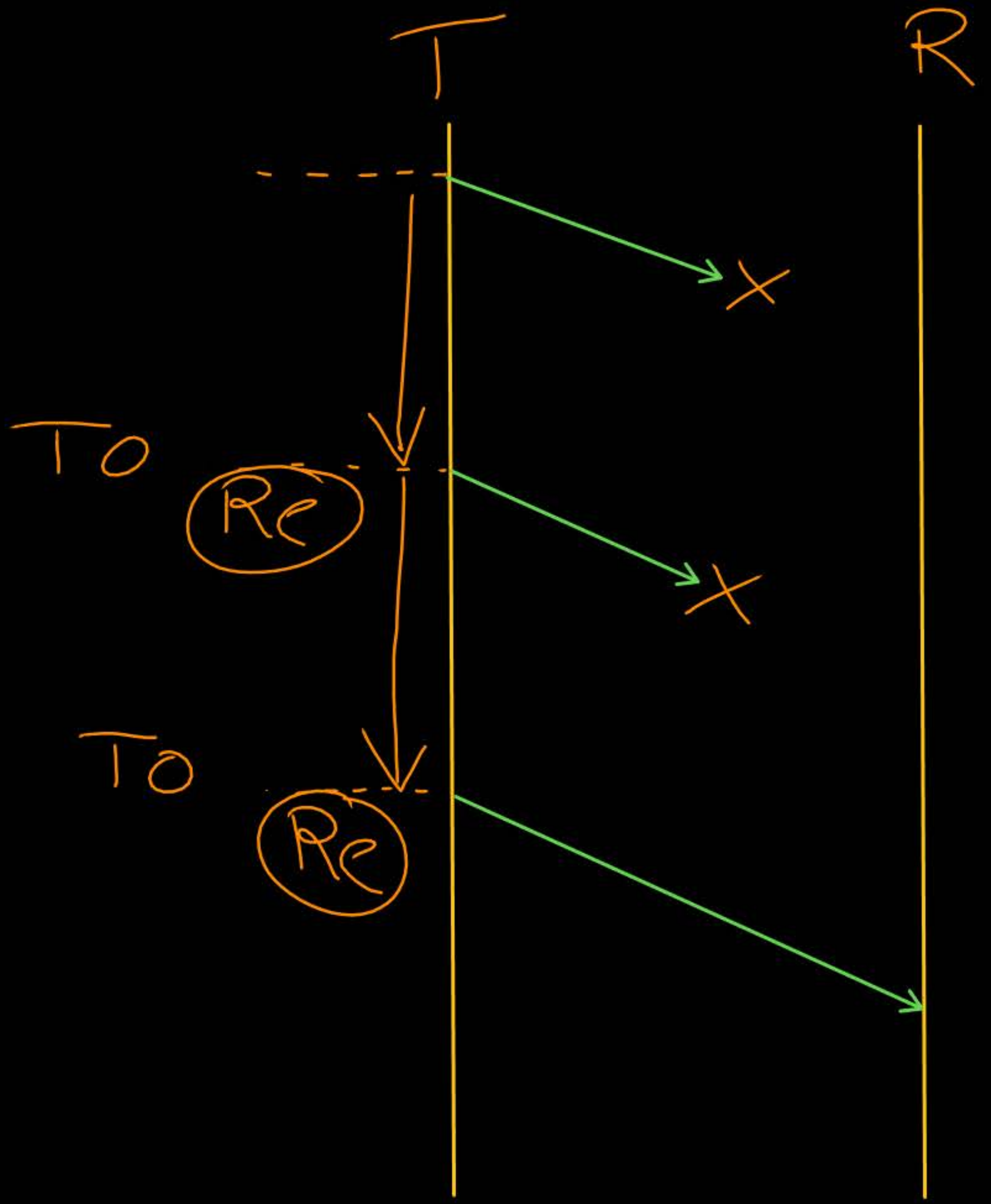
→ To avoid indefinite wait time at transmitter
transmitter uses "Automatic Repeat Request" (ARQ)

Automatic Repeat Request (ARQ) :

→ After transmission of a frame,
transmitter wait for an ACK upto time-out

→ After time-out, transmitter retransmit the frame





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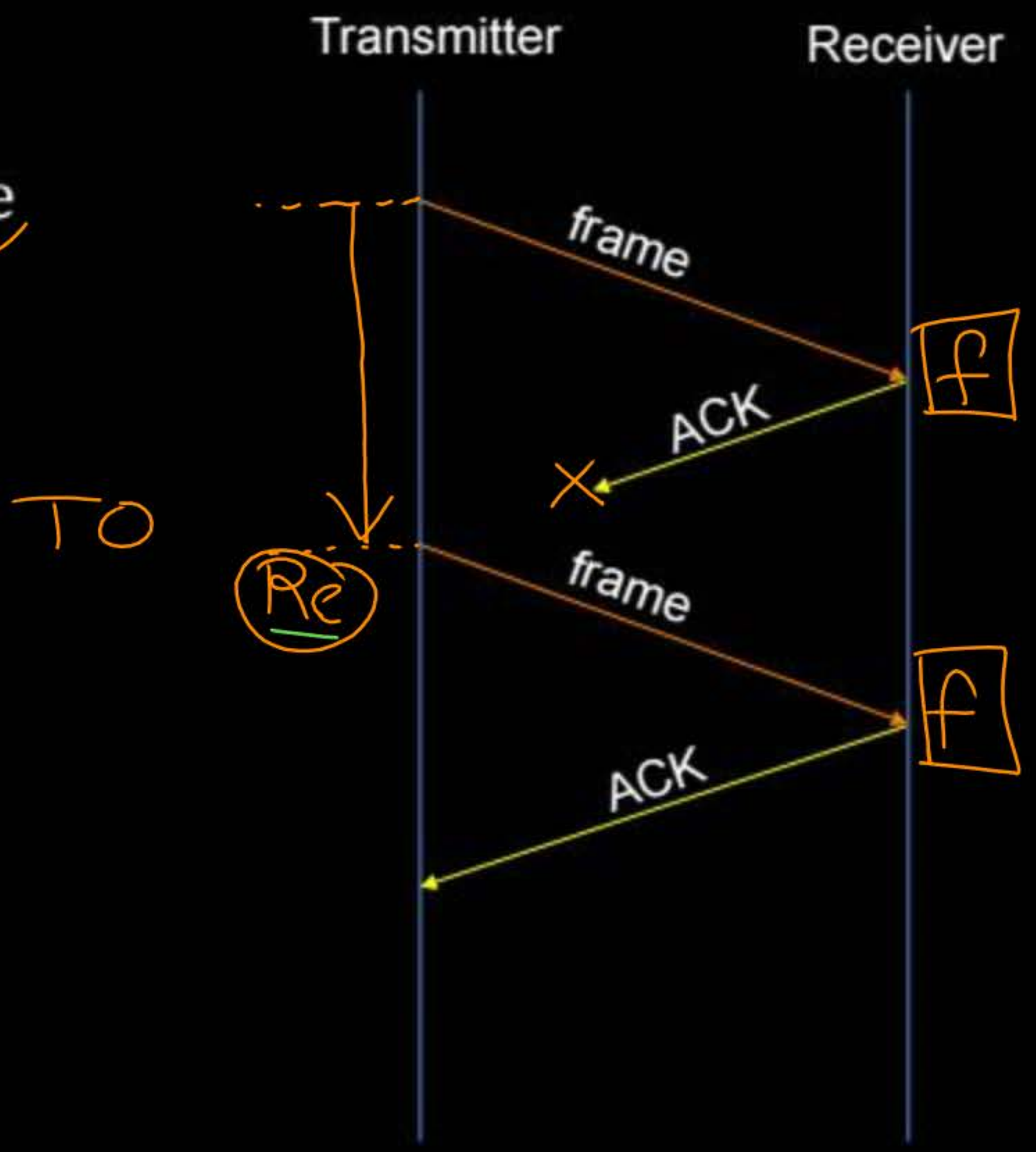
Topic : Stop-and-Wait ARQ

Case II :

→ Receiver may not able to identify duplicate frame



Topic : Stop-and-Wait ARQ

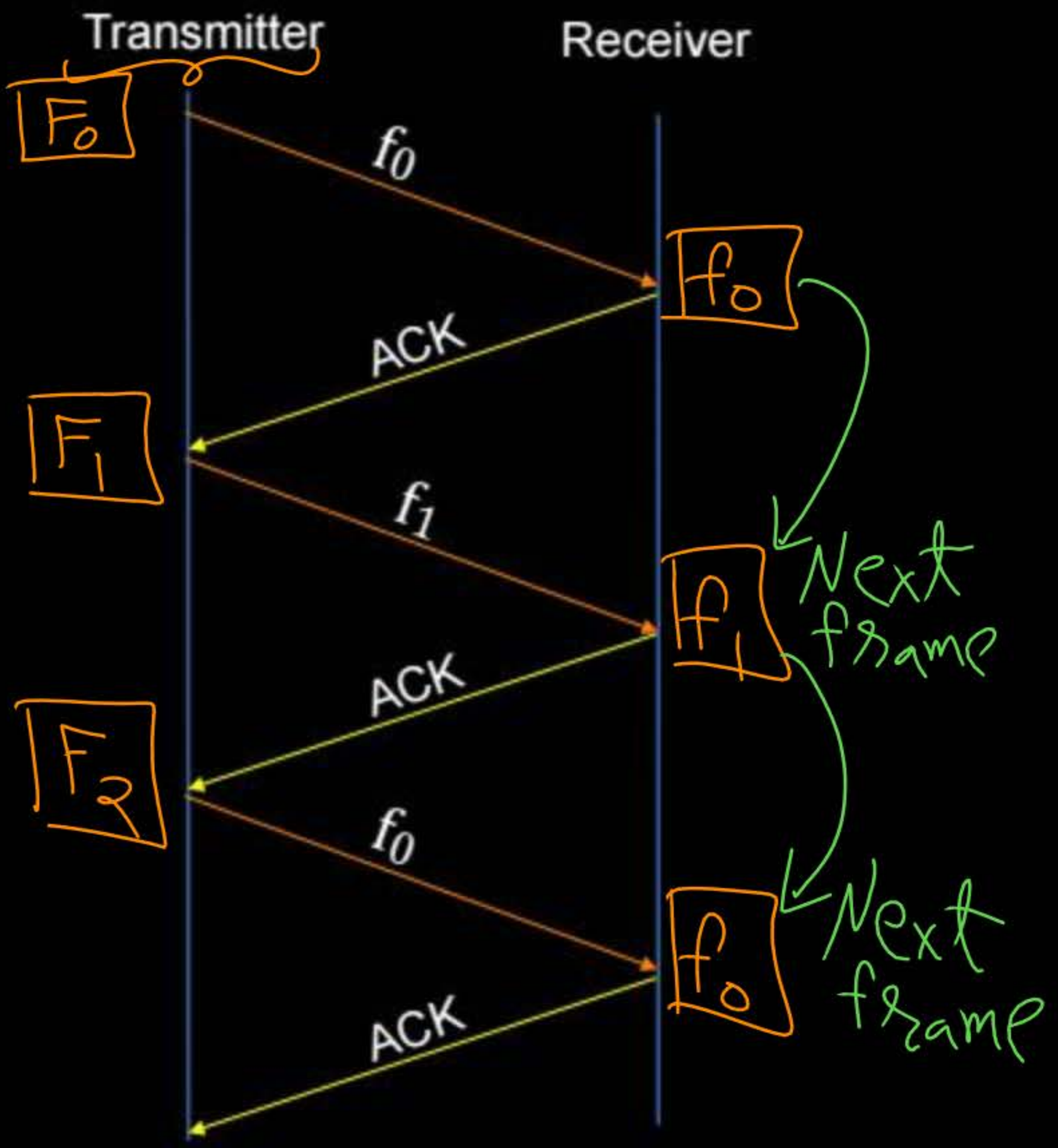
→ To identify duplicate frame at receiver transmitter uses "Sequence Number" field in the frame

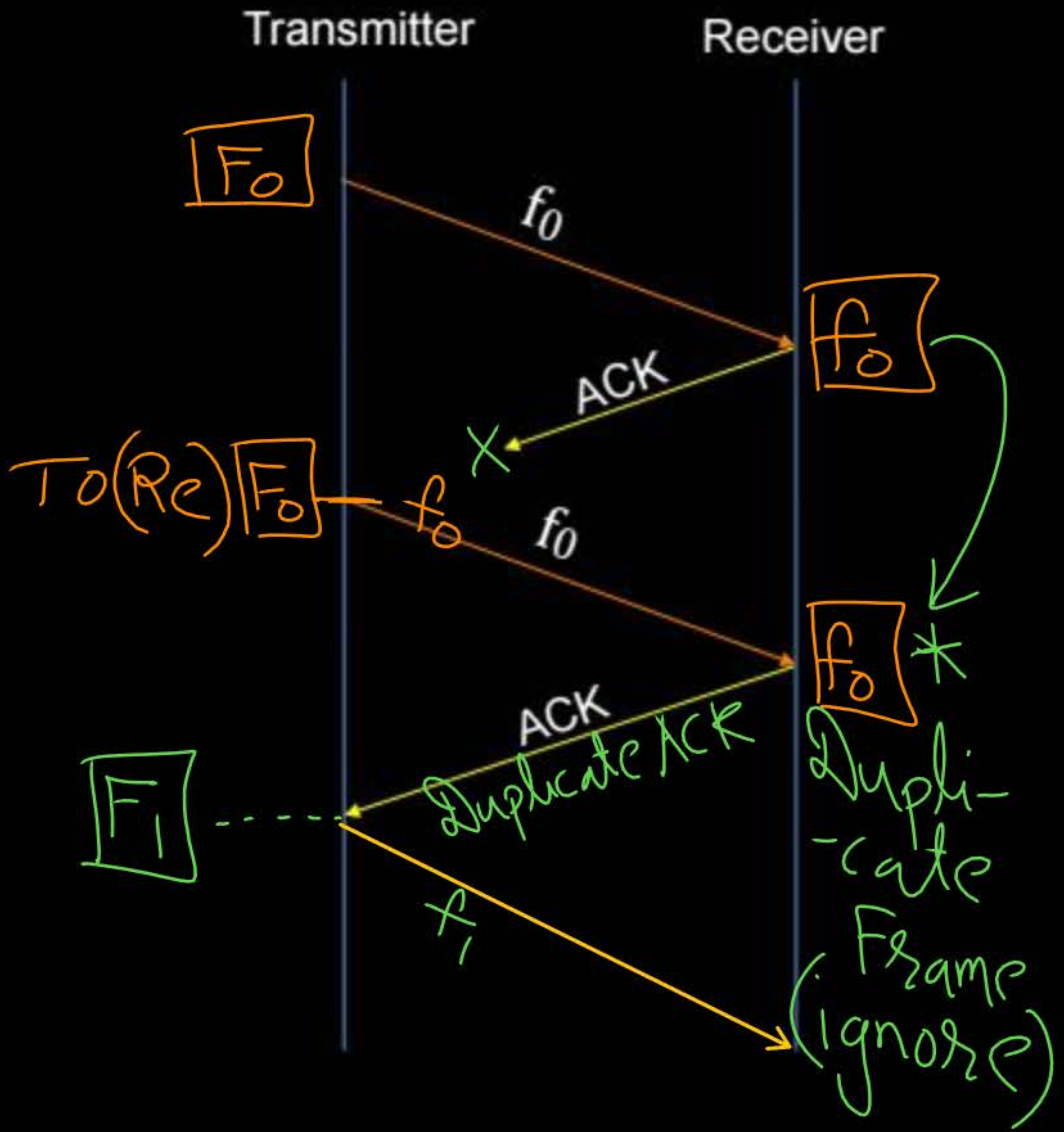
Sequence Number ← (Frame Number) mod 2

F_0	F_1	F_2	F_3	F_4	F_5	. . . . . .
f_0	f_1	f_0	f_1	f_0	f_1	

→ Stop-and-Wait ARQ is also known as "Alternate bit protocol"

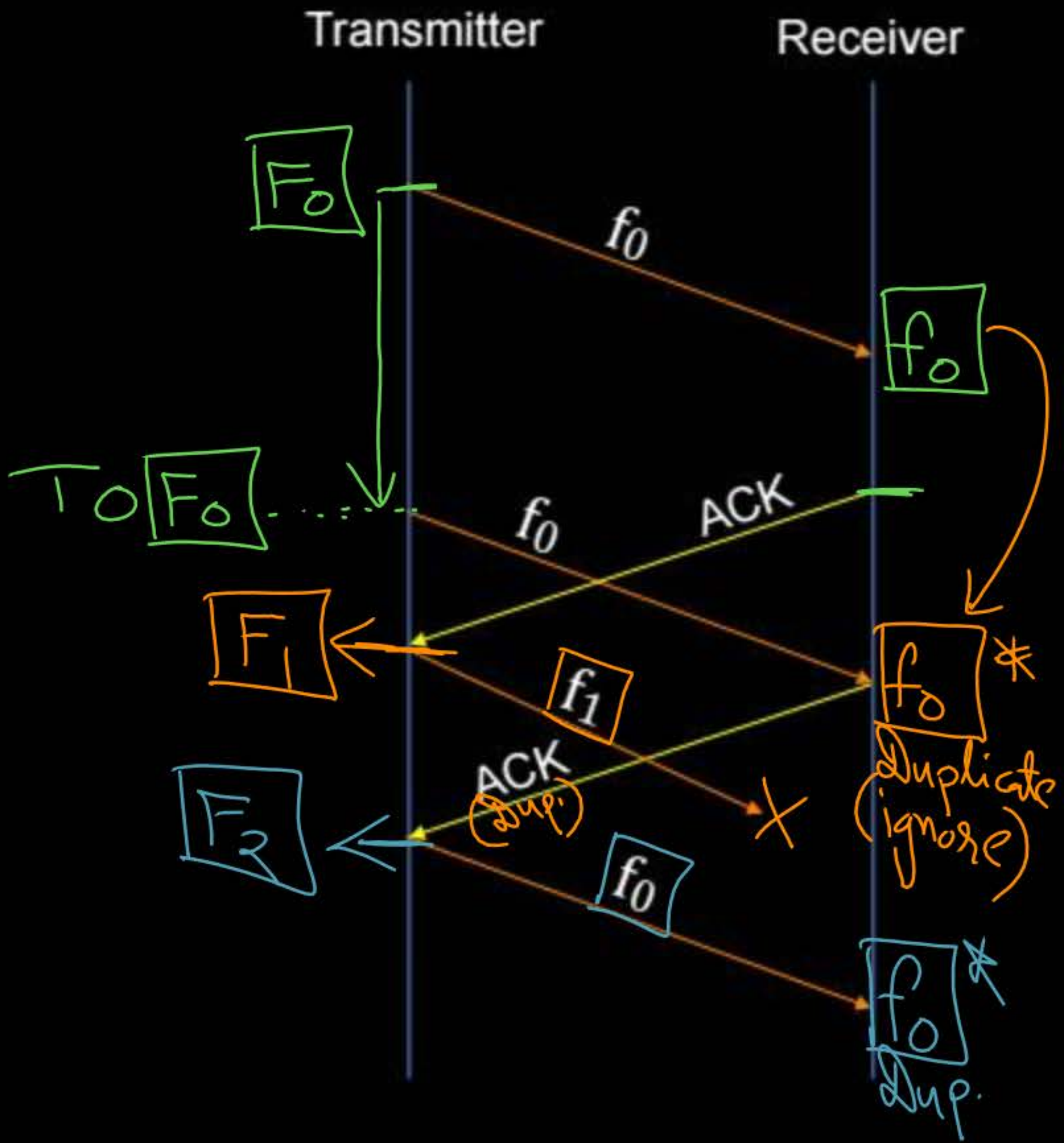
Topic : Stop-and-Wait ARQ





Case III :

→ Transmitter may not retransmit lost frame

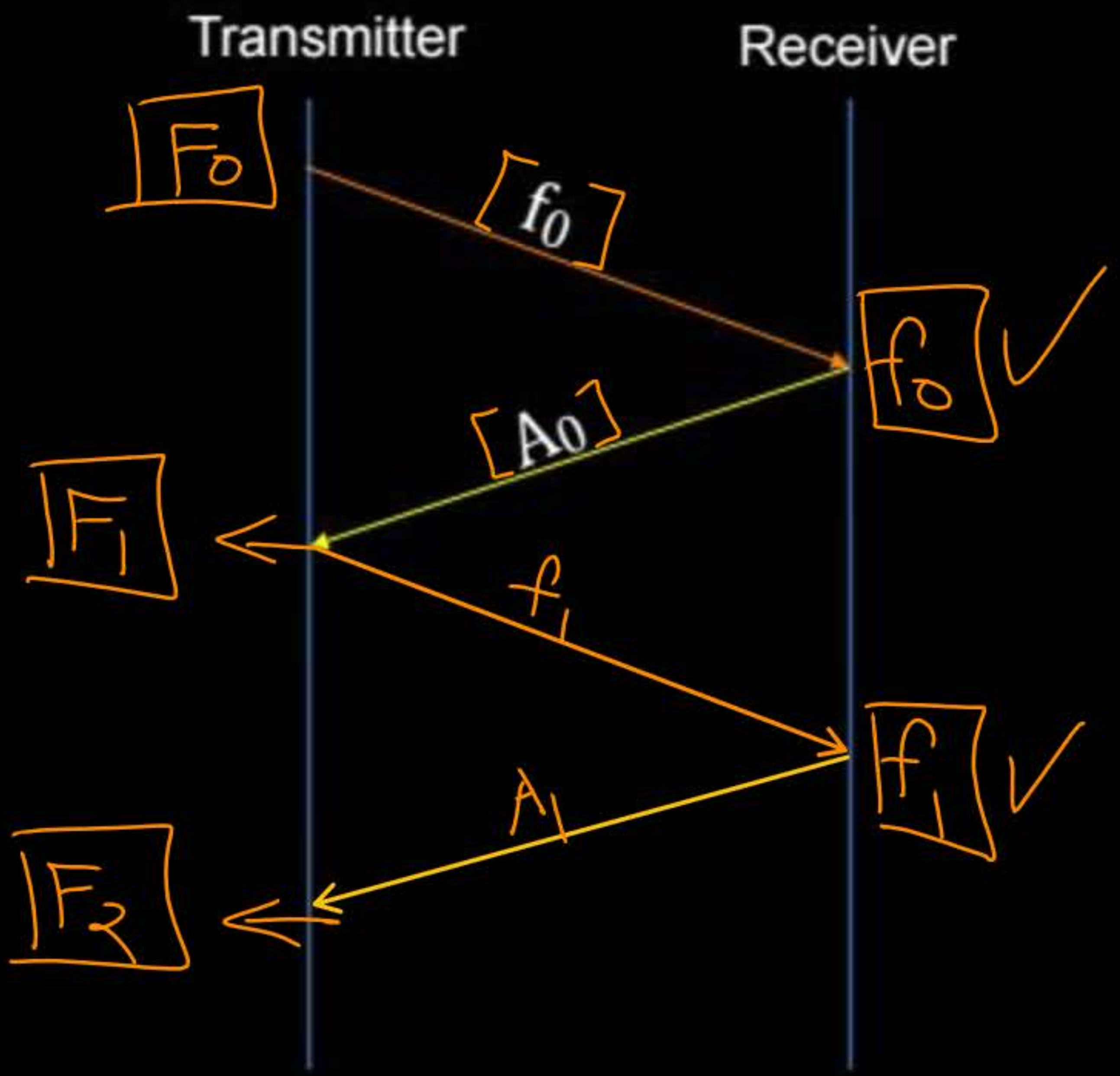


Topic : Stop-and-Wait ARQ

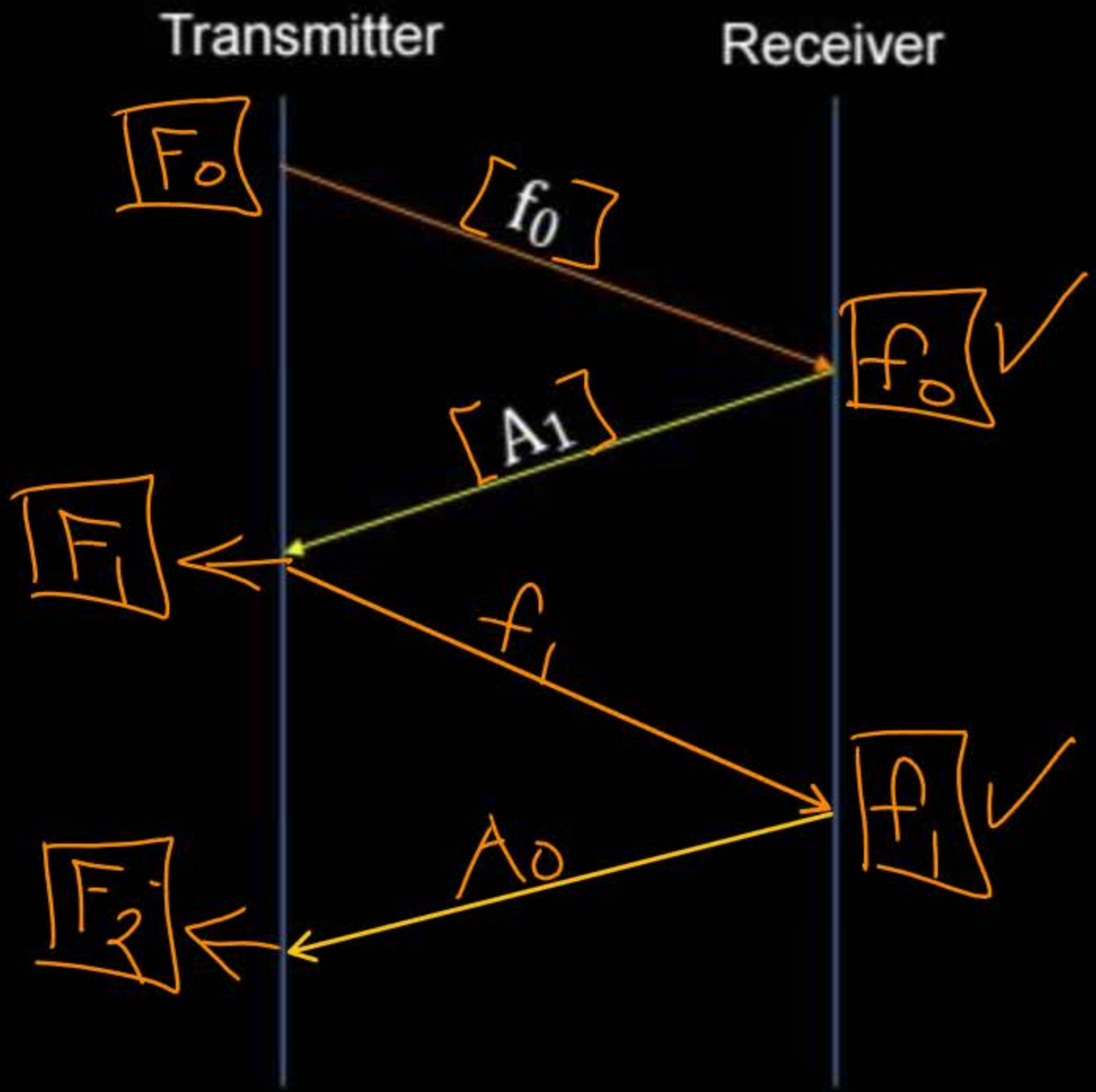
→ To ensure retransmission of every lost frame at transmitter receiver uses "Acknowledgment Number" in the ACK

Topic : Stop-and-Wait ARQ

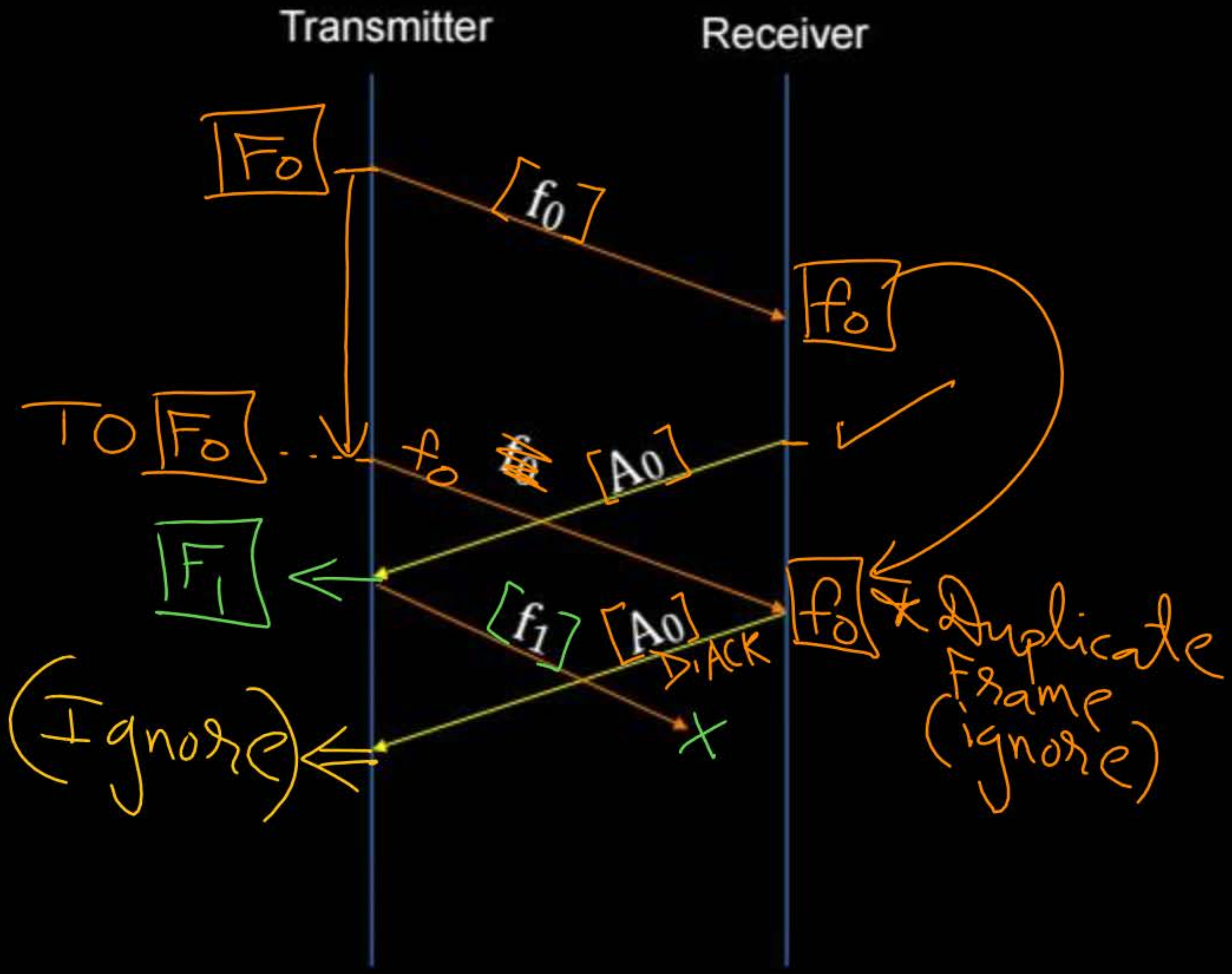
CASE I ✓



CASE II ⟹ TCP



Topic : Stop-and-Wait ARQ



2 mins Summary

Topic: Stop-and-Wait ARQ

THANK - YOU

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